



Abhishek Kumar Singh

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Examination	Degree	Institute	Year	CPI/%
Post Graduation (CSE)	MS By Research	IIT Bombay	2022-25	8.48
Graduation (CSE)	BTech	REC Sonbhadra	2018-22	8.13

WORK EXPERIENCE & COLLABORATION

- Qualcomm, Bengaluru India** *Open Source at- [Git-Hub Link](#)* *Machine Learning Engineer*
Team : Cloud ML (Model onboarding) | LLMs | Agents | Efficient Transformers *(July'25 - Present)*
 - Delivered **60% compile-time speedup** by adding subfunction support in Efficient Transformers.
 - Implemented **layer-wise execution** for large models (for example, Kimi K2.5), reducing peak memory to **91 GB RAM** and enabling export of a 1 trillion-parameter model under **1.5 hours; patent filing in progress**.
 - Performed a detailed analysis of **torch.compile** versus **torch.onnx.export** for model onboarding and deployment reliability.
 - Developed **DocQGenie**, a **multi-agent system** that autonomously understands codebases, extracts repository knowledge, and generate comprehensive technical documentation from GitHub Repositories (Hackathon Project).
- IBM Research, Bengaluru India** [*Published at : NAACL 2025*] *Paper Link* *Research Intern AI*
Team: Conversational AI | LLMs | RAG *(May'24 - Aug'24)*
 - Submitted **2 papers** during the internship and released a **large multilingual QA benchmark** to evaluate context-grounded question-answering in open-source multilingual LLMs.
 - Conducted a comparative study of **multilingual-query** and **translation-based** pipelines for Indian language QA systems.
 - Initiated a multi-LLM composition framework that combines model-specific strengths to improve multilingual QA performance (later extended as part of M.S. thesis work).

RESEARCH PROJECTS & PUBLICATIONS

- EIGEN: Expert-informed joint learning aggregation for high-fidelity information extraction from document images** [*Published at : ML4H (co-located with NeurIPS 2023)*] *Paper Link* *M.S. Thesis*
Guide: Prof. Ganesh Ramakrishnan | NER | Information Extraction | SPEAR | PyTorch *(Jan'23 - Jan'25)*
 - Developed **CAGE**, an unsupervised token classification approach that reduces dependency on labeled training data.
 - Fine-tuned **LayoutLM** as a strong baseline for document-level token classification tasks.
 - Built a **joint learning framework** using **SPEAR**, leveraging semi-supervised learning to achieve state-of-the-art results.
 - Proposed a **joint LLM fine-tuning strategy** with data programming, improving extraction accuracy while reducing labeling effort and cost.
 - Automated **labeling-function generation** and developed a deployable workflow suitable for practical usage.
- Composition of LLMs for Multilingual Question Answering** *M.S. Thesis*
Guide: Prof. Ganesh Ramakrishnan | Multilingual NLP | LLM Composition | RAG *(May'24 - Jan'25)*
 - Proposed and implemented a **multilingual LLM composition framework** that combines multiple specialized LLMs, leveraging their complementary linguistic and reasoning capabilities for question answering.
 - Implemented the complete end-to-end training and inference pipeline, including a **two-stage optimization strategy** for effective model composition and knowledge integration.
 - Demonstrated strong **zero-shot cross-lingual generalization**, achieving competitive performance on downstream tasks and languages **unseen during training**, while improving multilingual answer generation quality.

PROJECTS

- Fair Diffusion: Mitigating Bias in Conditional Diffusion Models** *Advanced Machine Learning*
Guide: Prof. Sunita Sarawagi | Generative Models | Debiasing | PyTorch *(Spring'23)*
 - Identified and mitigated bias issues in Conditional Denoising Diffusion Probabilistic Models.
 - Applied **supervised contrastive loss** to reduce bias in generated images.
 - Conducted extensive experiments on correlated and imbalanced MNIST datasets with quantified fairness metrics.

- **Fact-Checking Claims from Wikipedia** *Organization of Web Information (Spring'24)*
 Guide: **Prof. Soumen Chakrabarti** | RAG | Retrieval
 - Implemented an embedding-based retrieval system using **Sentence Transformer** for Wikipedia data, achieving a Recall@100 of **0.707**.
 - Enhanced retrieval precision by integrating **cross-encoder** re-ranking and **Named Entity Recognition (NER)**, reducing false positives by **28%** across claim types.
- **Reverse Codex: Automatic Code Comment Generation** *Speech, NLP, and the Web (Autumn'22)*
 Guide: **Prof. Pushpak Bhattacharyya** | Bi-LSTM | Sequence-to-Sequence | PyTorch
 - Developed an encoder-decoder model with attention for automatic Java code comment generation.
 - Implemented **Structured-Based Traversal (SBT)** for precise Abstract Syntax Tree (AST) navigation and code representation.
 - Achieved BLEU scores competitive with state-of-the-art models, demonstrating model efficacy.
- **OCR to Speech Pipeline: End-to-End Document-to-Audio System** *Deep Learning for NLP (Spring'23)*
 Guide: **Prof. Pushpak Bhattacharyya** | OCR | Summarization | Machine Translation | TTS
 - Fine-tuned pre-trained OCR models on the DocBank dataset and integrated PEGASUS for abstractive summarization.
 - Built transformer-based neural machine translation on the CFLIT Hindi dataset for accurate cross-lingual translation.
 - Implemented text-to-speech synthesis to complete the end-to-end document-to-audio pipeline.
- **Breast Cancer Classifier** *Foundation of Machine Learning (Autumn'22)*
 Guide: **Prof. Preethi Jyothi** | CNN | SVM | Ensemble | PCA | Keras
 - Conducted model comparisons across **SVM**, ensemble methods (**XGBoost**), and **CNN**.
 - Employed **PCA** for dimensionality reduction on the **BreakHis** dataset.
 - Determined CNN as the **optimal** performer.
- **Simulation of Mining Attacks on P2P Cryptocurrency network** *Introduction to Blockchains, Cryptocurrencies (Spring'24)*
 Guide: **Prof. Vinay Ribeiro** | Blockchain | Simulation
 - Designed a **discrete-event** simulator for a **peer-to-peer** cryptocurrency network for 10-100 users.
 - Simulated **selfish mining** scenarios with **dual attackers** to assess their impact on blockchain integrity.
 - Evaluated the influence of varying miner **hash power** and **network delays** on transaction throughput and efficiency.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, MATLAB, bash
- **Development:** HTML, CSS, PHP, GitHub Actions
- **Tools & Libraries:** Git, Scikit-learn, PyTorch, Keras, Hugging Face Transformers, ONNX, L^AT_EX

ACHIEVEMENTS AND EXTRA-CURRICULAR

- Achieved a **99.47** percentile (AIR 405) in GATE 2022.
- First author of a paper published in **ML4H 2023**, co-located with NeurIPS 2023, focusing on information extraction from document images.
- Achieved **Top Performer** in **TCS NQT 2022**.
- **Hobbies** : Cricket, Travelling, Chess, Music, Web Series